Kittiphun Ruttanakorn



Computer Technical Officer

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PROFESSIONAL SUMMARY

A computer technical officer with combined expertise in IT infrastructure and full-stack software development. I design, deploy, and maintain enterprise networks built on world-class equipment such as Cisco, administer Windows Server and Linux environments efficiently through the command-line interface, and build full-stack information systems that make the university's operations and services more reliable, secure, and modern. I also bring a background in industrial electronics and supervisory experience in manufacturing — leading electrical testing in R&D and machinery maintenance on production lines.

CORE COMPETENCIES

Enterprise Networking

Cisco IOS / IOS-XE CLI · Routing & Switching · VLAN · Trunking · STP · OSPF · Static Routing · ACL & Network Security · Wireless LAN · Monitoring & Troubleshooting

Windows Server

Active Directory (AD DS) · DNS · DHCP · Group Policy · File & Print Services · Hyper-V · IIS · PowerShell

Linux Server & CLI

Ubuntu · Debian · RHEL-based · Bash Scripting · SSH & Hardening · Nginx · Apache · Docker · systemd · cron · Log Analysis

Full-Stack Development

HTML · CSS · JavaScript · Node.js · Python · REST API · SQL Databases · Git & GitHub · IoT (ESP32) · VBA Automation

WORK EXPERIENCE

Computer Technical Officer

Oct 2024 – Present

Mahachulalongkornrajavidyalaya University, Phrae Campus

- Design, configure, and maintain the campus network on Cisco equipment for high stability and security.
- Administer Windows Server and Linux servers, including user accounts, system services, security, and backups.
- Develop information systems and digital tools that support administration, procurement, and academic work.
- Manage the university's digital-media public relations.
- Provide technical consultation and support to staff and students.

Freelance Full-Stack Developer

2019 – 2023

Self-employed

- Built full-stack web applications and information systems, covering the user interface, back end and database.
- Handled projects end to end — from gathering requirements and design through development to delivery.

Head of Machinery Maintenance 2017 – 2018

Thai An Pao Paper Products Co., Ltd. (Saha Group, Sriracha, Chonburi)

- Led maintenance of production machinery for paper cups and paper/plastic packaging.
- Production supplied packaging to Thai President Foods PCL.

Head of Electrical Testing Laboratory, Research & Development 2013 – 2017

Siam Compressor Industry Co., Ltd.

- Tested and developed projects on compressor packaging and installation under various operating conditions in refrigeration systems.
- Supervised the electrical testing laboratory within the R&D division.

KEY PROJECTS

Precision QC JavaScript · Python · WebAssembly

Image-based quality control for fungal cultures

8 analysis panels · 100% in-browser processing · ISO/IEC 17025 PDF report structure · 3 UI languages

- Loads Pyodide (WebAssembly) to run the Python engine directly in the browser — no server, and images never leave the device
- Image processing with NumPy + Pillow: detects the dish with a Hough transform and converts pixels to centimetres

kittiphun.rut.mcupr.ac.th/projects/precision-qc-web

Smart IV Alert C++ · ESP32 · IoT · KiCad

IoT system for monitoring and alerting IV infusions (Smart IV Alert)

28,000+ lines of C++ firmware · 8 beds per central unit · 1 s ESP-NOW reporting interval · 10/10 detector test scenarios passed

- Schematics and PCBs in KiCad 7, generated from the pin numbers in the firmware so drawings and code never drift apart
- 28,000+ lines of C++ firmware across the bedside unit, three central-unit display variants and a flow simulator
- ESP-NOW, Wi-Fi + WebServer, mDNS, and Preferences (NVS) for persistent settings

github.com/kittiphunr/IV-Monitor-BCNPH

Buddhist Art Gallery 360° JavaScript · Python · OpenCV · 360°

Street View-style 360° virtual tour · MCU Phrae Campus

28 360° viewpoints · 70 artworks with info cards · 276 image pairs matched with SIFT · 2–6 walkable directions per spot

- Built on Pannellum (WebGL): floor arrows aim themselves from map positions, return links are added automatically, and the heading carries over as you walk
- Matches SIFT features (OpenCV) and solves a spherical essential matrix to get the direction and rotation between every nearby pair of panoramas — 276 pairs
- A build script packs only the files the site uses (~27 MB out of ~720 MB of originals) and versions the CSS/JS URLs

kittiphun.rut.mcupr.ac.th/projects/bda-mcu-phrae

Other Projects

- **TOR-MCU-Standard** — A toolkit for preparing Terms of Reference (TOR) documents according to university standards. (VBA, Automation)
- **BOQ-MCU** — A tool for preparing Bills of Quantities (BOQ) and construction cost estimates. (Python)
- **MCUPR-CANTEEN** — Canteen management system for MCU Phrae Campus. (C++, Embedded)
- **Music Player ESP32-S3** — A music player with a 2.8-inch display built on the ESP32-S3 microcontroller. (C, ESP32, Embedded)

EDUCATION

Bachelor of Technology (B.Tech.) in Electronics Technology
Institute of Vocational Education: Northern Region 2

Vocational Certificate (Voc. Cert.) in Electronics
Phrae Technical College

Higher Vocational Certificate (High Voc. Cert.) in Electronics (Industrial Electronics)
Phrae Technical College